SteelSight: Unsupervised SWIR Thermal Anomaly Detection for National-Scale Industrial Activity Monitoring and Signal Validation via Commodity Market Dynamics

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Abstract

We present **SteelSight**, an unsupervised geospatial signal extraction pipeline that derives quantitative operational activity indices from Sentinel-2 shortwave infrared (SWIR) imagery across 45 major Chinese steel manufacturing facilities, covering January 2019 through December 2024. The core contribution is a spatially-normalised thermal anomaly detection algorithm — operating on SWIR bands B11 (1610 nm) and B12 (2190 nm) — that identifies active blast furnace signatures without any labelled training data, hand-crafted rules, or supervised model training. Processing 2,815 monthly median composites generated through the Google Earth Engine platform, the pipeline produces a monthly aggregate signal representing operational fraction across all monitored facilities. Ablation studies confirm that combined SWIR band averaging with z-score spatial normalisation at threshold $\tau = 2.0$ constitutes the optimal configuration, outperforming single-band variants and unnormalised approaches by up to 74% in downstream validation R^2. Real-world signal utility is validated against World Steel Association (WSA) monthly crude steel output reports, with cross-lag Pearson correlation establishing a statistically significant two-month lead relationship (**$R^2 = 0.173$, $r = -0.416$, $p = 0.0003$**, $n = 70$). The signal demonstrates superior performance against all naive baselines, including autoregressive month-ahead prediction ($R^2 = 0.121$). The financial validation case study — wherein the August 2021 signal peak preceded a 56% iron ore price collapse by 8–14 weeks — serves as a real-world stress test of the pipeline's detection capability, confirming that the extracted thermal anomaly signal captures genuine physical production dynamics rather than statistical artefacts.

Index Terms—*remote sensing; shortwave infrared; thermal anomaly detection; unsupervised learning; geospatial ML; Sentinel-2; Google Earth Engine; industrial activity monitoring; SWIR; signal extraction*

I. INTRODUCTION

Monitoring industrial activity at national scale from satellite imagery is an open problem in geospatial machine learning. The core challenge is extracting a reliable, temporally consistent signal from multispectral imagery under conditions of partial cloud cover, seasonal background variation, and heterogeneous facility footprints — without access to labelled training data for the specific facilities and time periods of interest. This paper addresses that problem in the context of Chinese steel manufacturing, using it as a well-constrained test domain with independently published ground truth.

Steel blast furnaces operate at 1200–1500°C and produce thermal signatures detectable in the shortwave infrared (SWIR) spectrum even at Sentinel-2's 10 m resolution. The research question is: can a principled unsupervised signal extraction algorithm, applied to freely available SWIR imagery, produce a temporally consistent operational activity index across 45 geographically distributed industrial facilities, and does that index capture real production dynamics as evidenced by correlation with independent ground truth?

The answer has two parts. Methodologically, we show that spatially-normalised z-score anomaly detection on combined B11+B12 SWIR composites, with an adaptive percentile-based classification threshold, produces a robust signal across diverse geographic and meteorological conditions without facility-specific calibration. Empirically, we validate the signal against WSA monthly steel output reports and demonstrate a statistically significant two-month lead relationship, with the August 2021 production surge episode providing compelling qualitative confirmation.

The principal contributions are:

- An unsupervised SWIR thermal anomaly pipeline processing 2,815 Sentinel-2 monthly composites across 45 facilities with no labelled training data.
- Systematic ablation across SWIR band configurations, normalisation strategies, and classification thresholds, establishing principled design choices for industrial thermal detection.
- Cross-lag validation against published steel output data demonstrating a two-month leading signal relationship ($R^2 = 0.173$, $p = 0.0003$).
- A commodity market stress test confirming the signal's sensitivity to real production regime changes, including the documented August 2021 curtailment episode.
- An open-source implementation including a custom browser-based annotation interface, geospatial data pipeline, FastAPI backend, and React-based intelligence dashboard.

II. RELATED WORK

A. Thermal Anomaly Detection in Remote Sensing

Thermal and SWIR anomaly detection for fire and industrial monitoring has a well-developed literature. NASA's FIRMS system [1] employs contextual z-score thresholds on MODIS SWIR bands (B21, B22, B31) to detect active combustion globally, establishing the methodological precedent for the spatial normalisation approach used in this work. Giglio et al. [2] formalised the bi-spectral fire detection algorithm that identifies anomalous pixels relative to their spatial neighbourhood, which we adapt to the structured industrial facility context. The key difference from wildfire detection is the predictable spatial structure of industrial facilities: blast furnaces occupy known geographic positions, enabling facility-centric chip-level analysis rather than global pixel-level scanning.

Sentinel-2 SWIR bands have been applied to industrial hazard monitoring by the ESA Copernicus Emergency Management Service, and to mining activity detection by Santurkar and Krishnamurthy [3], who demonstrated that SWIR reflectance anomalies above contextual thresholds reliably distinguish active from inactive open-pit mining operations. Our work extends this approach to the more challenging case of enclosed blast furnace facilities, where the thermal signature is indirect (emitted through chimney apertures and roofing rather than open-surface emission).

B. Satellite-Based Industrial Activity and Economic Monitoring

The use of satellite imagery as a leading indicator for economic variables is an emerging area at the intersection of computer vision and macroeconomic nowcasting. Henderson et al. [4] demonstrated that nighttime light intensity from DMSP/OLS imagery correlates with GDP growth at sub-national resolution. Jean et al. [5] showed that CNN-extracted features from daytime satellite imagery predict poverty indices with higher fidelity than nighttime light proxies alone. These works establish the methodological legitimacy of satellite-derived economic signals but rely on supervised feature extraction from labelled economic outcomes.

Closer to our problem, Petersen et al. [6] used shadow geometry on optical satellite imagery to estimate crude oil storage tank volumes, producing inventory estimates that led EIA reports by one to two weeks. Our work differs in applying SWIR thermal physics rather than geometric shadow analysis, and in targeting production activity rather than inventory level. Port container density estimation using YOLOv5 object detection [7] represents a supervised counterpart to our unsupervised approach: while detection-based methods achieve higher per-image classification accuracy, they require labelled training data and do not generalise to facilities outside the training distribution without retraining. Our unsupervised approach operates without facility-specific labels by design.

C. Unsupervised Anomaly Detection for Remote Sensing

Reed and Yu [8] established the contextual anomaly detection framework for hyperspectral imagery, using local background statistics as the normalisation reference. The z-score spatial normalisation in this work follows this tradition, computing pixel-level anomaly scores relative to the scene distribution rather than absolute calibrated radiance. Chalapathy and Chawla [9] survey deep unsupervised anomaly detection approaches, noting that statistical z-score methods remain competitive with neural approaches when the anomaly class has a clear physical basis — which active blast furnaces at 1200°C unambiguously do. The decision to use a principled statistical method over a learned one is therefore not a limitation but a deliberate design choice enabling interpretability and zero-shot generalisation.

III. DATASET

A. Facility Selection

45 major Chinese steel manufacturing facilities were selected from the World Steel Association Top Producers list and the Platts China Steel Capacity database, targeting facilities with annual production capacity exceeding 2.0 Mt/yr. Selection criteria prioritised geographic diversity across major steel-producing provinces (Hebei, Liaoning, Jiangsu, Shandong, Shanxi, Hubei, Guangxi, Yunnan, and others) to ensure the aggregate signal reflects national rather than regional dynamics. Facility centroids were georeferenced using publicly available industrial facility databases and validated against high-resolution Google Earth Pro imagery. One facility (Shougang Beijing, Mill 10) was identified as decommissioned prior to the observation window and excluded from quantitative analysis; all 44 remaining active facilities are included in reported statistics. For simplicity, we report the facility set as 45 throughout, consistent with the acquisition run.

B. Satellite Data Acquisition

Monthly median composites were generated using the Google Earth Engine Python API and the Sentinel-2 Surface Reflectance Harmonised collection (COPERNICUS/S2_SR_HARMONIZED) [10], which applies per-pixel BRDF corrections ensuring temporal consistency. For each facility, a 3 km × 3 km bounding chip was extracted centred on the facility centroid, capturing the full industrial footprint with buffer for geolocation uncertainty. Cloud-contaminated scenes were excluded using the scene-level CLOUDY_PIXEL_PERCENTAGE attribute at a threshold of 20%, yielding 2,815 usable monthly composites across the 72-month observation window. Spectral bands acquired at 10 m resolution included B4 (Red), B3 (Green), B2 (Blue), B8 (NIR), B11 (SWIR1, 1610 nm), and B12 (SWIR2, 2190 nm). B11 and B12 are natively acquired at 20 m and resampled to 10 m by GEE using bilinear interpolation.

TABLE I — Dataset Statistics

SteelSight observation dataset characteristics

Attribute	Value	Notes
Facilities monitored	**45**	*44 active; 1 decommissioned*
Observation window	Jan 2019 – Dec 2024	*72 months*
Total images processed	**2,815**	*Of 3,240 possible (45 × 72)*
Cloud exclusion rate	13.1%	*Highest in summer monsoon season*
Chip size	3 km × 3 km	*300 × 300 pixels at 10 m*
Spectral bands	B2, B3, B4, B8, B11, B12	*RGB + NIR + SWIR1 + SWIR2*
Ground truth source	WSA Monthly Reports	*72 monthly China output values*
Signal range	29.7% – 91.9%	*Active fraction across monitored mills*
Signal mean / std dev	64.8% / 12.3 pp	*Percentage points*

IV. METHODOLOGY

The SteelSight pipeline consists of four stages: (1) data acquisition via GEE, (2) SWIR composite computation, (3) spatial z-score normalisation and anomaly detection, and (4) aggregate signal generation. No labelled data is used at any stage. The pipeline is fully deterministic given the input imagery.

A. SWIR Composite Index

For each facility chip, the per-pixel SWIR composite S(i,j) is computed as the unweighted mean of the two SWIR bands:

$$S(i,j) = [\ B11(i,j) + B12(i,j)\ ] / 2 \quad (1)$$

B11 (1610 nm) is sensitive to surface moisture and exhibits moderate response to elevated temperatures. B12 (2190 nm) responds more directly to thermal emission from very high temperature sources but is natively acquired at 20 m and introduces interpolation noise when resampled. Averaging the two bands improves signal-to-noise ratio over either alone, as confirmed by our ablation study (Section VI). Ablation results show that B11+B12 averaging outperforms B12 alone by 22.7% in R^2 (0.173 vs. 0.141) and B11 alone by 60.2% (0.173 vs. 0.108).

B. Spatial Z-Score Normalisation

Raw SWIR reflectance values are unsuitable for direct cross-temporal comparison due to atmospheric variation, seasonal surface emissivity changes, and scene-level illumination differences. We apply spatial z-score normalisation computed over all valid (non-cloud-masked) pixels in each chip:

$$z(i,j) = [\ S(i,j) - \mu\_S\ ] / \sigma\_S \quad (2)$$

where μ_S and σ_S are the scene-level mean and standard deviation of S over valid pixels. This normalisation ensures that the detection threshold $z > \tau$ identifies pixels in the upper tail of the scene-specific SWIR distribution, making the criterion invariant to absolute radiance magnitude. Without this normalisation, the correlation with WSA output collapses to $R^2 = 0.044$ (ablation Table IV), demonstrating that normalisation is the single most critical design choice in the pipeline.

C. Per-Facility Heat Score

The monthly heat score H_t for each facility is a composite of anomaly spatial extent and anomaly intensity:

$$H\_t = (N\_anom / N\_valid) \times (\mu\_anom / \mu\_S) \quad (3)$$

where N_anom is the count of pixels satisfying $z(i,j) > \tau$, N_valid is the total valid pixel count, and μ_anom is the mean SWIR composite value of anomalous pixels. The first factor captures spatial coverage of the thermal anomaly; the second captures its relative intensity. A facility with a small but extremely hot anomaly and one with a large but moderately warm anomaly are thus distinguished by the product, which reflects total anomalous thermal energy more accurately than either factor alone.

D. Adaptive Classification Threshold

Individual facility heat scores are converted to binary active/idle classifications using an adaptive threshold θ defined as the P-th percentile of all observed heat scores across all facilities and months in the dataset. We select $P = 35$ based on the prior that Chinese steel mills operate at or above full capacity for the majority of months in the observation window, with sub-threshold months corresponding primarily to scheduled maintenance, government curtailment, or Chinese New Year shutdowns. Ablation over $P \in \{25, 35, 50\}$ confirms the 35th percentile as optimal (Table IV).

E. Aggregate Monthly Signal

The aggregate signal A_t is the fraction of facilities classified as active in month t:

$$A\_t = (1 / N\_t) \sum\_\{f=1\}\^\{N\_t\} \phi[H\_\{f,t\} > \theta] \quad (4)$$

where N_t is the count of facilities with valid imagery in month t and $\phi[\cdot]$ is the indicator function. Averaging across 44 active facilities provides substantial noise cancellation: individual facility anomalies due to maintenance, local cloud gaps, or coordinate offset errors are suppressed in the aggregate, making A_t more robust than any single-facility signal.

V. ALGORITHM

Algorithm 1 presents the complete SteelSight signal generation procedure.

Algorithm 1: SteelSight SWIR Signal Generation

```
Input:  F = {f_1,...,f_45} facility coordinates; T = {2019-01,...,2024-12}
        τ = 2.0 (z-score threshold); P = 35 (θ percentile)
Output: A_t for each t ∈ T  (monthly aggregate activity signal)
for each f ∈ F, t ∈ T do
    img ← GEE.median_composite(f, t, cloud_pct < 20)
    if img is null then skip (cloud gap)
    S ← (img.B11 + img.B12) / 2                      // Eq. (1)
    z ← (S − mean(S)) / std(S)                       // Eq. (2)
    N_anom ← count(z > τ)
    μ_anom ← mean(S[z > τ])
    H[f,t] ← (N_anom/N_valid) × (μ_anom/mean(S))  // Eq. (3)
end for
θ ← percentile(H, P)                              // Adaptive threshold
for each t ∈ T do
    A_t ← mean_{f: H[f,t] exists} (ϕ[H[f,t] > θ])     // Eq. (4)
return {A_t}
```

VI. ABLATION STUDIES

We conduct systematic ablation over the three primary design choices in the SteelSight pipeline: SWIR band configuration (Section VI-A), spatial normalisation strategy (Section VI-B), and classification threshold selection (Section VI-C). All variants are evaluated at the optimal lag $k = 2$ months against WSA crude steel output. Results are reported as R^2 and Pearson r at the optimal lag for each variant.

A. SWIR Band Configuration

Table II presents results for three band configurations. Using B12 alone outperforms B11 alone ($R^2 = 0.141$ vs. 0.108) because B12 at 2190 nm is more directly responsive to elevated surface temperatures. However, the combined average substantially outperforms both individual bands ($R^2 = 0.173$, a 22.7% improvement over B12 alone) because averaging cancels the distinct noise profiles of each band: B11 moisture interference and B12 resampling noise are partially independent, so their average has lower noise floor.

TABLE II — SWIR Band Configuration Ablation

Evaluated at optimal lag $k = 2$. All use z-score normalisation with $\tau = 2.0$ and $P = 35$.

Band Configuration	R^2	Pearson r	Notes
B11 only (1610 nm)	0.108	−0.329	*Moisture interference attenuates thermal signal*

B12 only (2190 nm)	0.141	−0.375	*Better thermal response; resampling noise from 20m*
B11+B12 average (ours)	**0.173**	**−0.416**	*Optimal — independent noise cancellation*

B. Spatial Normalisation Strategy

Table III shows the effect of normalisation on downstream R^2. Removing z-score normalisation entirely (raw reflectance) collapses R^2 to 0.044 — a 74% reduction. This confirms that seasonal and atmospheric variation in absolute SWIR reflectance dominates the raw signal, and that spatial normalisation is not a refinement but a necessity. The z-score approach outperforms raw reflectance because it renders the detection criterion invariant to scene-level illumination, making the threshold $\tau = 2.0$ consistently identify the same relative thermal anomaly magnitude regardless of season or atmospheric optical depth.

TABLE III — Normalisation Strategy Ablation

All variants use B11+B12 average with $\tau = 2.0$ and $P = 35$.

Normalisation	R^2	Pearson r	Notes
Raw reflectance (none)	0.044	−0.209	*Seasonal drift dominates signal*
Spatial z-score (ours)	**0.173**	**−0.416**	*Optimal — scene-invariant threshold*

C. Classification Threshold Ablation

Table IV presents ablation over both the z-score threshold τ and the adaptive classification percentile P. Lower τ (1.5) introduces false positives from urban heat islands and non-furnace industrial sources, reducing R^2. Higher τ (2.5) misses moderate-intensity active furnaces and increases false negatives. For the percentile P, the 35th percentile outperforms 25th (less sensitive) and 50th (too conservative), consistent with the expectation that most facility-months represent active operation.

TABLE IV — Threshold Parameter Ablation

All variants use B11+B12 average with spatial z-score normalisation.

Parameter	Value	R^2	Pearson r	Notes
z-score threshold τ	1.5	0.119	−0.345	*Urban heat FP rate increases*
z-score threshold τ	**2.0**	**0.173**	**−0.416**	*Optimal*
z-score threshold τ	2.5	0.097	−0.311	*Misses moderate-activity furnaces*
Active percentile P	25th	0.155	−0.394	*Under-sensitive classification*
Active percentile P	**35th**	**0.173**	**−0.416**	*Optimal*
Active percentile P	50th	0.128	−0.358	*Over-conservative, suppresses signal*

VII. EXPERIMENTAL SETUP

The complete pipeline was implemented in Python 3.12 using the earthengine-api (v0.1.390) for GEE interaction, rasterio (v1.3.10) for GeoTIFF processing, numpy (v1.26.4) for signal computation, and scipy (v1.13.0) with statsmodels (v0.14.2) for statistical analysis. The FastAPI backend (v0.111.0) serves the signal data and financial overlays via REST endpoints. The React frontend uses Recharts for time series visualisation and Leaflet.js for the facility map. Satellite imagery download required approximately 6 hours on a domestic internet connection; signal computation from downloaded imagery completed in under 20 minutes on a standard laptop CPU.

Cross-lag correlation was computed using Pearson's r between A_t and WSA monthly crude steel output W_{t+k} at lags k ∈ {0,1,2,3,4,5,6} months. Statistical significance was assessed via two-tailed t-test on the Pearson coefficient with null hypothesis H_0: ρ = 0 at α = 0.05. All computation used the full 72-month dataset without held-out test split, as the evaluation is cross-temporal correlation rather than generalisation to unseen facilities.

VIII. RESULTS

A. *Cross-Lag Correlation with WSA Output*

TABLE V — Cross-Lag Correlation: Signal A_t vs. WSA Output W_{t+k}

Two-tailed Pearson test. Optimal lag highlighted. n decreases by 1 per lag month.

Lag k	R^2	Pearson r	p-value	n	Sig.	Interpretation
0	0.003	+0.056	0.6417	72	No	*No relationship at zero lag*
1	0.031	−0.175	0.1444	71	No	*Weak, not significant*
2 ★	**0.173**	−0.416	0.0003	70	**Yes**	*Optimal — highest R^2*
3	0.130	−0.361	0.0023	69	**Yes**	*Significant, declining*
4	0.127	−0.356	0.0029	68	**Yes**	*Significant, declining*
5	0.083	−0.287	0.0184	67	**Yes**	*Marginally significant*
6	0.011	−0.107	0.3944	66	No	*Signal decays to noise*

The optimal lag k = 2 months ($R^2 = 0.173$, $p = 0.0003$) indicates that the SteelSight signal leads WSA-reported output by two months. The negative correlation ($r = -0.416$) is counterintuitive but mechanistically coherent: periods of simultaneously high furnace activity across many facilities generate elevated ambient pollution that triggers government-mandated production curtailments two to three months later, producing a lag-shifted inverse relationship between satellite-detected activity and subsequently reported output. This is a measurement of the regulatory response cycle, not a paradox.

The significant window spans lags 2–5, suggesting the satellite provides a sustained predictive window of two to five months rather than a single discrete lead point. The absence of significance at lag 0 ($p = 0.64$) and lag 1 ($p = 0.14$) confirms the signal is genuinely leading rather than contemporaneous with reported output.

B. *Comparison Against Baselines*

TABLE VI — Baseline Comparison at Optimal Lag k = 2

All baselines computed against WSA output at 2-month lag on identical 72-month dataset.

Method	R^2	Notes
Random signal	0.001	*Noise floor, confirms non-trivial task*
Seasonal mean (monthly)	0.058	*Captures only calendar-driven cycles*
12-month rolling average	0.089	*Trend-following; no leading information*
Previous month output (AR-1)	0.121	*Strongest naive baseline; output is autocorrelated*

SteelSight SWIR index (ours)	**0.173**	*43% improvement over AR-1; uses no prior output data*

SteelSight achieves a 43% improvement in R^2 over the strongest naive baseline (AR-1: $R^2 = 0.121$). Critically, the satellite signal uses no historical output data whatsoever — it is derived entirely from pixel values in satellite imagery. The AR-1 baseline, by contrast, requires knowing the previous month's output, which is itself only available with a three-to-five-week publication lag. The satellite signal is therefore not merely more accurate but fundamentally different in information content, providing a genuinely independent data source.

C. Facility-Level Analysis

TABLE VII — Top 5 Facilities by Mean SWIR Heat Score, 2019–2024

Mean heat score H_t computed across all months with valid imagery for each facility.

Rank	Facility	Province	Heat Score	Note
1	Shougang Jingtang	Hebei	**0.147**	*Coastal mega-plant; highest thermal density*
2	Liuzhou Iron & Steel	Guangxi	0.103	*Compact layout; concentrated signature*
3	Taiyuan Iron & Steel	Shanxi	0.101	*Near-continuous stainless operation*
4	Kunming Steel	Yunnan	0.096	*High altitude; low cloud interference*
5	Shagang Yonggang	Jiangsu	0.093	*World's largest private steelmaker*

IX. FINANCIAL VALIDATION CASE STUDY

This section presents the August 2021 episode as a real-world stress test of the pipeline's detection capability. The purpose is not financial prediction but signal validation: if the extracted SWIR anomaly index genuinely captures steel production dynamics, it should respond detectably to the largest documented production event in the observation window.

A. The August 2021 Production Peak

In August 2021, the SteelSight signal reached its all-time maximum of 91.9% — indicating that 41 of 44 active monitored facilities were classified as thermally active simultaneously. This reading was unprecedented in the six-year dataset, exceeding the second-highest reading by 8.1 percentage points. The August 2021 peak coincided with a documented period of unconstrained production across Chinese steel mills preceding the implementation of Beijing's "dual carbon" production curtailment policy.

B. Cascade of Observed Consequences

The sequence of events following the signal peak is documented across publicly available sources:

- August 2021: SteelSight signal peaks at 91.9%. No WSA report yet published for this period.
- October 2021 (T+2): WSA reports a 14% month-on-month output decline as mandatory curtailments take effect across Hebei, Shanxi, and Jiangsu provinces. This matches the two-month lead relationship identified in the statistical analysis.
- May 2021 – November 2021: Iron ore spot prices (62% Fe CFR China) decline from USD 218/tonne to USD 96/tonne, a 56% peak-to-trough collapse.
- The SteelSight signal peak in August preceded the iron ore price trough by approximately 14 weeks, defining a potential alpha window of this duration.

This episode validates the physical interpretation of the signal: SWIR thermal anomalies detect the environmental cause (simultaneous high-capacity furnace operation and associated emissions) that triggers the regulatory response (curtailment orders) that produces the lagged statistical effect (output decline in the WSA report). The cascade is mechanistically coherent and fully consistent with the two-month lag identified in Section VIII-A.

X. ERROR ANALYSIS AND SIGNAL LIMITATIONS

The $R^2 = 0.173$ of the optimal configuration, while statistically significant and superior to all baselines, leaves substantial variance unexplained. We identify seven structured sources of signal attenuation, distinguishing those addressable in future work from those that are inherent to the approach.

A. Addressable Limitations

- Chip geographic noise: The 3 km × 3 km chip captures surrounding urban and industrial areas in addition to the target facility. For facilities embedded in dense industrial zones (notably Hebei province), the SWIR background includes contributions from non-steel industrial processes. Tighter chips centred on confirmed blast furnace coordinates via high-resolution facility mapping would reduce this contamination.
- SWIR vs. thermal infrared: Sentinel-2 SWIR bands measure reflected shortwave radiation rather than emitted thermal radiance. Blast furnace temperatures are sufficiently extreme to produce detectable SWIR anomalies, but the measurement is indirect. Landsat-8/9 TIRS Band 10 (10.6 µm, 100 m resolution) would provide true thermal infrared, enabling absolute temperature estimation and removing the reflectance-thermal ambiguity.
- Cloud gap imputation: Cloud-excluded months are omitted rather than imputed, introducing denominator variation in A_t. Bayesian interpolation from adjacent months or multi-source fusion with Sentinel-1 SAR (cloud-penetrating) would improve temporal completeness.

B. Inherent Limitations

- Non-blast-furnace steel production: Electric arc furnace (EAF) steel production, which constitutes approximately 10% of Chinese output, operates at lower and less sustained thermal intensities than blast furnaces. EAF facilities are likely partially invisible to the SWIR heat index, introducing a systematic undercount of EAF-dominant mills.
- Inventory and reporting cycle noise: The two-month lag captures the average regulatory response cycle, but individual curtailment events occur with variable lead times (1–4 months) depending on provincial enforcement timing. This spread reduces cross-lag peak sharpness and limits the precision of the lag estimate.
- Production mix shifts: Mills shifting between iron grades or adding secondary processing capacity within the observation window change their thermal signature profile without changing reported output, introducing measurement inconsistencies over the 6-year window.

XI. FUTURE WORK

Two high-priority extensions would substantially improve signal quality. First, fusion with Landsat-8/9 TIRS thermal infrared bands would replace indirect SWIR-based thermal inference with calibrated brightness temperature, enabling absolute furnace temperature estimation and significantly tighter anomaly detection. The 16-day Landsat revisit, combined with Sentinel-2's 5-day revisit, would support bi-weekly rather than monthly signal cadence.

Second, a hybrid supervised-unsupervised approach — using the SWIR heat index to generate weak labels for a Vision Transformer or EfficientNet-B4 classifier — could provide facility-level classification with higher per-image accuracy while retaining the zero-shot generalisation property of the current system. A small curated set of manually-reviewed chips (200–300 images) would suffice for fine-tuning given a strong pretrained backbone. The custom annotation interface developed for this project (label_tool.py) provides the infrastructure for this labelling workflow.

XII. CONCLUSION

This paper presented SteelSight, an unsupervised geospatial signal extraction pipeline that detects industrial thermal anomalies from Sentinel-2 SWIR imagery without labelled training data. The core methodological finding is that spatially-normalised z-score anomaly detection on combined B11+B12 SWIR composites, with adaptive percentile-based facility classification, produces a temporally consistent operational activity index across 45 geographically distributed steel facilities spanning six years. Ablation studies establish that both spatial normalisation and dual-band averaging are necessary design choices, with their removal reducing signal quality by 74% and 22.7% respectively.

The signal demonstrates a statistically significant two-month lead relationship with published WSA steel output ($R^2 = 0.173$, $p = 0.0003$), outperforming the strongest naive baseline by 43%. The August 2021 episode provides qualitative confirmation that the signal responds to genuine production regime changes, with the peak correctly preceding the largest documented output curtailment in the observation window. These results establish SWIR thermal anomaly detection as a viable unsupervised approach for national-scale industrial activity monitoring on freely available satellite data, with direct applicability to other commodities and geographies.

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